

LAB 7

Fashion Management System

Objective

You are required to develop a backend application for managing **fashion clothing items** and **customer reviews**, allowing users to register, log in/out, manage products, and provide reviews. The system will include **full CRUD operations**, **JWT authentication**, and **Mongoose populate()** to display related data.

System Overview

The backend application manages **three main collections** in MongoDB:

- **Users** – account and authentication information
- **Clothes** – clothing items data
- **Reviews** – user ratings and comments

1. Users Collection

Field	Type	Description
_id	ObjectId	Unique identifier
username	String	Must be unique
password	String	Encrypted using bcrypt
role	String	Default: "user" (optional: "admin")
createdAt	Date	Timestamp when user registered

Example:

```
{
  "_id": "674b2a90b82af93a2b17a0f1",
  "username": "anna_fashion",
  "password": "$2b$10$34fghkl...",
  "role": "user",
  "createdAt": "2025-10-29T09:00:00Z"
}
```

2. Clothes Collection

Field	Type	Description
_id	ObjectId	Unique identifier
name	String	Clothing name
category	String	Dress, Jacket, T-shirt, Pants, etc.
brand	String	Fashion brand
price	Number	Price in USD
description	String	Product details
imageUrl	String	URL of product image
averageRating	Number	Calculated dynamically
createdAt	Date	Timestamp when added

3. Reviews Collection

Field	Type	Description
_id	ObjectId	Unique identifier
userId	ObjectId	Ref → Users
clothId	ObjectId	Ref → Clothes
rating	Number	Rating (1–5)
comment	String	Review text
createdAt	Date	Date/time of submission

Relationships

- A **User** can post many **Reviews**.
- A **Cloth item** can have many **Reviews**.
- Reviews link **Users** ↔ **Clothes** via `userId` and `clothId`.

General Requirements

- Database: `FashionDB_2025`
- Folder name: `FashionSys_StudentCode`
- Tools: Node.js, Express.js, MongoDB, Mongoose, JWT, bcrypt, dotenv, EJS.
- A `/resources` folder is provided with:
 - Sample JSON data
 - List of required npm packages
 - Example EJS template for rendering product detail page
 - Postman collection for API testing

Tasks to be Completed

Task 1 – Project Initialization & Configuration

1. Create a Node.js project named **FashionSys_StudentCode**.
2. Initialize with `npm init` and install dependencies:
3. `express mongoose bcrypt jsonwebtoken dotenv cors ejs`
4. Create `.env` file:
5. `PORT=1234`
6. `MONGO_URI=your_mongodb_connection`
7. `JWT_SECRET=mysecretkey`
8. Set up MongoDB database `FashionDB_2025` with three collections: **Users**, **Clothes**, **Reviews**.
9. Import sample JSON data or create your own.

Task 2 – User Authentication APIs

2.1 POST /users/register

- Registers a new user (`username, password`).
- Password must be hashed using **bcrypt**.
- Return success/failure message in JSON.

2.2 POST /users/login

- Validates user credentials.

- Returns a **JWT token** upon successful login.

Example response:

```
{
  "message": "Login successful",
  "token": "eyJhbGciOi..."
}
```

2.3 POST /users/logout

- (Frontend simulation) Clear token or respond with logout confirmation.
- Example JSON:

```
{ "message": "User logged out successfully." }
```

Task 3 – Clothes Management (CRUD APIs)

These endpoints manage clothing items in the collection.

Method	Endpoint	Description	Auth
GET	/clothes	Get all clothing items	Public
GET	/clothes/:id	Get details by ID	Public
POST	/clothes	Add new clothing item	Auth (admin)
PUT	/clothes/:id	Update clothing item by ID	Auth (admin)
DELETE	/clothes/:id	Delete clothing item	Auth (admin)

Example Create Request:

```
{
  "name": "Cotton T-shirt",
  "category": "T-shirt",
  "brand": "Uniqlo",
  "price": 19.99,
  "description": "Soft and breathable cotton tee.",
  "imageUrl": "/images/tshirt.jpg"
}
```

Example Get All Response:

```
[
  {
    "name": "Denim Jacket",
    "brand": "Levi's",
    "price": 120,
    "averageRating": 4.3
  },
  {
    "name": "Summer Dress",
    "brand": "Zara",
    "price": 79.99,
    "averageRating": 4.5
  }
]
```

Task 4 – Review Management (Authenticated Users)

Method	Endpoint	Description
POST	/reviews/:clothId	Add a new review for a clothing item
GET	/reviews/myreviews	View all reviews of the logged-in user
PUT	/reviews/:id	Update a review (only the owner can edit)

DELETE	/reviews/:id	Delete a review (only the owner can delete)
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- When creating or updating a review, recalculate `averageRating` in the corresponding **Clothes** document.
- Use Mongoose `.populate('clothId', 'name brand category')` to include related data.

Example GET /reviews/myreviews Response:

```
{
  "user": "anna_fashion",
  "reviews": [
    {
      "cloth": { "name": "Denim Jacket", "brand": "Levi's" },
      "rating": 5,
      "comment": "Stylish and comfy!",
      "createdAt": "2025-10-29T10:00:00Z"
    }
  ]
}
```

Task 5 – Advanced Data Display Using populate()

5.1 GET /clothes/details/:id (Authenticated)

- Returns full details of a clothing item, including **all reviews** and **reviewer usernames**.
- Use **nested populate()**:

```
Cloth.findById(req.params.id)
  .populate({
    path: 'reviews',
    populate: { path: 'userId', select: 'username' }
  });
```

Example Response:

```
{
  "_id": "674b2ae4f91a4a6210c74c20",
  "name": "Denim Jacket",
  "brand": "Levi's",
  "price": 120,
  "averageRating": 4.3,
  "reviews": [
    {
      "rating": 5,
      "comment": "Love it!",
      "user": { "username": "john_style" }
    }
  ]
}
```

5.2 GET /users/details/:id (Admin Only)

- Retrieve full user information, including all reviews written by the user and the clothes they reviewed.
- Use double populate:

```
User.findById(req.params.id)
  .populate({
    path: 'reviews',
    populate: { path: 'clothId', select: 'name brand category' }
  });
```

Task 6 – Optional: Render EJS View

- Create a simple EJS page (`views/cloth-details.ejs`) to render details from `/clothes/details/:id` API.
- Include:
 - Product image, name, price, description
 - List of all reviews (username + comment + rating)

Security & Authorization

- All sensitive routes (CRUD, reviews, etc.) require JWT verification:
- `Authorization: Bearer <token>`
- Admin role required for modifying clothes collection.

Expected Learning Outcomes

After completing this lab, students will:

1. Understand how to build full CRUD APIs using Express + MongoDB.
2. Implement secure JWT authentication and authorization.
3. Apply data relationships and populate in MongoDB.
4. Use EJS for rendering API data visually.
5. Build a realistic backend for an e-commerce or fashion system.

✓ Deliverables

- Node.js project folder `FashionSys_StudentCode`
- `.env` configuration
- Postman collection with all endpoints
- MongoDB Compass screenshot showing collections
- Optional: Screenshot of rendered EJS view

JSON DATA

1. users.json

```
[
  {
    "_id": { "$oid": "674b1001alf9a1c001" },
    "username": "anna_fashion",
    "password": "$2b$10$hashedpassword1",
    "role": "user",
    "createdAt": "2025-10-29T08:00:00Z"
  },
  {
    "_id": { "$oid": "674b1001alf9a1c002" },
    "username": "john_style",
    "password": "$2b$10$hashedpassword2",
    "role": "user",
    "createdAt": "2025-10-29T08:10:00Z"
  },
  {
    "_id": { "$oid": "674b1001alf9a1c003" },
    "username": "maria_look",
    "password": "$2b$10$hashedpassword3",
    "role": "user",
    "createdAt": "2025-10-29T08:20:00Z"
  },
  {
    "_id": { "$oid": "674b1001alf9a1c004" },
    "username": "david_trend",
    "password": "$2b$10$hashedpassword4",
    "role": "user",
    "createdAt": "2025-10-29T08:30:00Z"
  },
  {
    "_id": { "$oid": "674b1001alf9a1c005" },
    "username": "lucas_modern",
    "password": "$2b$10$hashedpassword5",
    "role": "user",
    "createdAt": "2025-10-29T08:40:00Z"
  },
  {
    "_id": { "$oid": "674b1001alf9a1c006" },
    "username": "jane_vogue",
    "password": "$2b$10$hashedpassword6",
    "role": "user",
    "createdAt": "2025-10-29T08:50:00Z"
  },
  {
    "_id": { "$oid": "674b1001alf9a1c007" },
    "username": "chris_lux",
    "password": "$2b$10$hashedpassword7",
    "role": "user",
    "createdAt": "2025-10-29T09:00:00Z"
  },
  {
    "_id": { "$oid": "674b1001alf9a1c008" },
    "username": "emily_glam",
    "password": "$2b$10$hashedpassword8",
    "role": "user",
    "createdAt": "2025-10-29T09:10:00Z"
  },
  {
    "_id": { "$oid": "674b1001alf9a1c009" },
    "username": "admin_fashion",
    "password": "$2b$10$hashedpassword9",
    "role": "admin",
  }
```

```

    "createdAt": "2025-10-29T09:20:00Z"
  },
  {
    "_id": { "$oid": "674b1001a1f9a1c010" },
    "username": "sophia_style",
    "password": "$2b$10$hashedpassword10",
    "role": "user",
    "createdAt": "2025-10-29T09:30:00Z"
  }
]

```

2. clothes.json

```

[
  {
    "_id": { "$oid": "674b2001b2f9a2c001" },
    "name": "Denim Jacket",
    "category": "Jacket",
    "brand": "Levi's",
    "price": 120,
    "description": "Classic blue denim jacket for all seasons.",
    "imageUrl": "/images/jacket01.jpg",
    "averageRating": 4.3
  },
  {
    "_id": { "$oid": "674b2001b2f9a2c002" },
    "name": "Summer Linen Dress",
    "category": "Dress",
    "brand": "Zara",
    "price": 79.99,
    "description": "Lightweight linen dress with modern cut.",
    "imageUrl": "/images/dress01.jpg",
    "averageRating": 4.6
  },
  {
    "_id": { "$oid": "674b2001b2f9a2c003" },
    "name": "Cotton T-shirt",
    "category": "T-shirt",
    "brand": "Uniqlo",
    "price": 19.99,
    "description": "Soft cotton T-shirt with relaxed fit.",
    "imageUrl": "/images/tshirt01.jpg",
    "averageRating": 4.4
  },
  {
    "_id": { "$oid": "674b2001b2f9a2c004" },
    "name": "Leather Boots",
    "category": "Shoes",
    "brand": "H&M",
    "price": 89.99,
    "description": "Durable leather boots for urban fashion.",
    "imageUrl": "/images/boots01.jpg",
    "averageRating": 4.2
  },
  {
    "_id": { "$oid": "674b2001b2f9a2c005" },
    "name": "Slim Fit Jeans",
    "category": "Pants",
    "brand": "Levi's",
    "price": 69.99,
    "description": "Stylish slim-fit jeans with stretch comfort.",
    "imageUrl": "/images/jeans01.jpg",
    "averageRating": 4.5
  },
  {
    "_id": { "$oid": "674b2001b2f9a2c006" },
    "name": "Wool Sweater",

```

```

    "category": "Sweater",
    "brand": "Mango",
    "price": 59.99,
    "description": "Warm wool sweater for cool weather.",
    "imageUrl": "/images/sweater01.jpg",
    "averageRating": 4.7
  },
  {
    "_id": { "$oid": "674b2001b2f9a2c007" },
    "name": "Sports Hoodie",
    "category": "Hoodie",
    "brand": "Nike",
    "price": 65.00,
    "description": "Comfortable hoodie suitable for workouts.",
    "imageUrl": "/images/hoodie01.jpg",
    "averageRating": 4.8
  },
  {
    "_id": { "$oid": "674b2001b2f9a2c008" },
    "name": "Silk Scarf",
    "category": "Accessory",
    "brand": "Gucci",
    "price": 150,
    "description": "Elegant silk scarf for special occasions.",
    "imageUrl": "/images/scarf01.jpg",
    "averageRating": 4.9
  },
  {
    "_id": { "$oid": "674b2001b2f9a2c009" },
    "name": "Classic Blazer",
    "category": "Blazer",
    "brand": "Hugo Boss",
    "price": 210,
    "description": "Tailored blazer suitable for formal wear.",
    "imageUrl": "/images/blazer01.jpg",
    "averageRating": 4.5
  },
  {
    "_id": { "$oid": "674b2001b2f9a2c010" },
    "name": "Running Shorts",
    "category": "Sportswear",
    "brand": "Adidas",
    "price": 35,
    "description": "Lightweight and breathable running shorts.",
    "imageUrl": "/images/shorts01.jpg",
    "averageRating": 4.1
  }
]

```

3. reviews.json

Mỗi review liên kết `userId` và `clothId` để minh họa quan hệ populate.

```

[
  {
    "_id": { "$oid": "674b3001c3f9a3c001" },
    "userId": { "$oid": "674b1001a1f9a1c001" },
    "clothId": { "$oid": "674b2001b2f9a2c001" },
    "rating": 5,
    "comment": "Perfect denim jacket - fits great!",
    "createdAt": "2025-10-29T10:00:00Z"
  },
  {
    "_id": { "$oid": "674b3001c3f9a3c002" },
    "userId": { "$oid": "674b1001a1f9a1c002" },
    "clothId": { "$oid": "674b2001b2f9a2c002" },

```



```

    "rating": 4,
    "comment": "Nice material, a bit long though.",
    "createdAt": "2025-10-29T10:05:00Z"
  },
  {
    "_id": { "$oid": "674b3001c3f9a3c003" },
    "userId": { "$oid": "674b1001a1f9a1c003" },
    "clothId": { "$oid": "674b2001b2f9a2c003" },
    "rating": 5,
    "comment": "The T-shirt is very soft and comfortable.",
    "createdAt": "2025-10-29T10:10:00Z"
  },
  {
    "_id": { "$oid": "674b3001c3f9a3c004" },
    "userId": { "$oid": "674b1001a1f9a1c004" },
    "clothId": { "$oid": "674b2001b2f9a2c004" },
    "rating": 4,
    "comment": "Good quality leather boots!",
    "createdAt": "2025-10-29T10:15:00Z"
  },
  {
    "_id": { "$oid": "674b3001c3f9a3c005" },
    "userId": { "$oid": "674b1001a1f9a1c005" },
    "clothId": { "$oid": "674b2001b2f9a2c005" },
    "rating": 5,
    "comment": "Great jeans with perfect fit!",
    "createdAt": "2025-10-29T10:20:00Z"
  },
  {
    "_id": { "$oid": "674b3001c3f9a3c006" },
    "userId": { "$oid": "674b1001a1f9a1c006" },
    "clothId": { "$oid": "674b2001b2f9a2c006" },
    "rating": 5,
    "comment": "So warm and cozy sweater.",
    "createdAt": "2025-10-29T10:25:00Z"
  },
  {
    "_id": { "$oid": "674b3001c3f9a3c007" },
    "userId": { "$oid": "674b1001a1f9a1c007" },
    "clothId": { "$oid": "674b2001b2f9a2c007" },
    "rating": 4,
    "comment": "Nice hoodie but color fades slightly.",
    "createdAt": "2025-10-29T10:30:00Z"
  },
  {
    "_id": { "$oid": "674b3001c3f9a3c008" },
    "userId": { "$oid": "674b1001a1f9a1c008" },
    "clothId": { "$oid": "674b2001b2f9a2c008" },
    "rating": 5,
    "comment": "Absolutely luxurious scarf!",
    "createdAt": "2025-10-29T10:35:00Z"
  },
  {
    "_id": { "$oid": "674b3001c3f9a3c009" },
    "userId": { "$oid": "674b1001a1f9a1c009" },
    "clothId": { "$oid": "674b2001b2f9a2c009" },
    "rating": 5,
    "comment": "Perfect blazer for business meetings.",
    "createdAt": "2025-10-29T10:40:00Z"
  },
  {
    "_id": { "$oid": "674b3001c3f9a3c010" },
    "userId": { "$oid": "674b1001a1f9a1c010" },
    "clothId": { "$oid": "674b2001b2f9a2c010" },
    "rating": 4,
    "comment": "Light and comfortable shorts for daily jogging.",
    "createdAt": "2025-10-29T10:45:00Z"
  }

```

```
}  
]
```

☒ Tóm tắt

Collection	Số lượng	Quan hệ
users	10	1 user có thể có nhiều reviews
clothes	10	1 item có thể có nhiều reviews
reviews	10	mỗi review nối <code>userId</code> ↔ <code>clothId</code>